9K Generate the Last-to-First Mapping of a String

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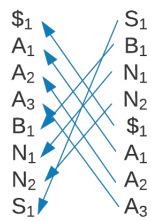
Last-to-First Mapping Problem
Construct a Last-to-First Array of a Burrows-Wheeler Transform.

Input: A string *Transform* representing the Burrows-Wheeler transform of an unknown string *Text*.
Output: An array *LastToFirst* that provides the following information: given a symbol at position i in *LastColumn* of the Burrows-Wheeler matrix of *Text*, LASTTOFIRST(i) reveals its position in *FirstColumn*.



Formatting

Input: A string *Transform* representing the Burrows-Wheeler transform of an unknown string *Text*.
Output: A space-separated list of integers representing the array *LastToFirst*.

Constraints

- The length of *Transform* will be between 1 and 10^3.
- The value of i will be between 1 and 10^3.

Test Cases

Case 1

Description: The sample dataset is not actually run on your code.

Input:
```
smnpbnaaaaa$a
```

Output:
```
13 8 9 12 7 10 11 1 2 3 4 5 0 6
```

Case 2

Description: There are no repeat characters in *Text*.

Input:
```
T$ACG
```

Output:
```
4 0 1 2 3
```

Case 3

Description: *Text* is made up of only one character.

Input:
```
AAAAAAAAAA$
```

Output:
```
1 2 3 4 5 6 7 8 9 10 0
```

Case 4

Description: *Text* is palindromic or has substrings that are palindromic.

Input:
```
TGCG$AA
```

Output:
```
5 0 3 2 6 1 4
```

Case 5

Description: A larger dataset of the same size as that provided by the randomized autograder.